

ECE 3441 Fall 2021

Exam 2

Version B

Please do not open booklet unless instructed to do so, or substantial penalty will result.

- ? Closed book, closed notes
- ? ONE 8.5 x 11" double-sided cheat sheet is allowed.
- ? Only four function calculators are allowed.
- ? Show all details of your work to obtain credit (The answer alone will not suffice).
- ? You may not talk during the quiz to others or to yourself.
- ? You have 80 minutes.

First Name Last Name,

Peoplesoft ID,

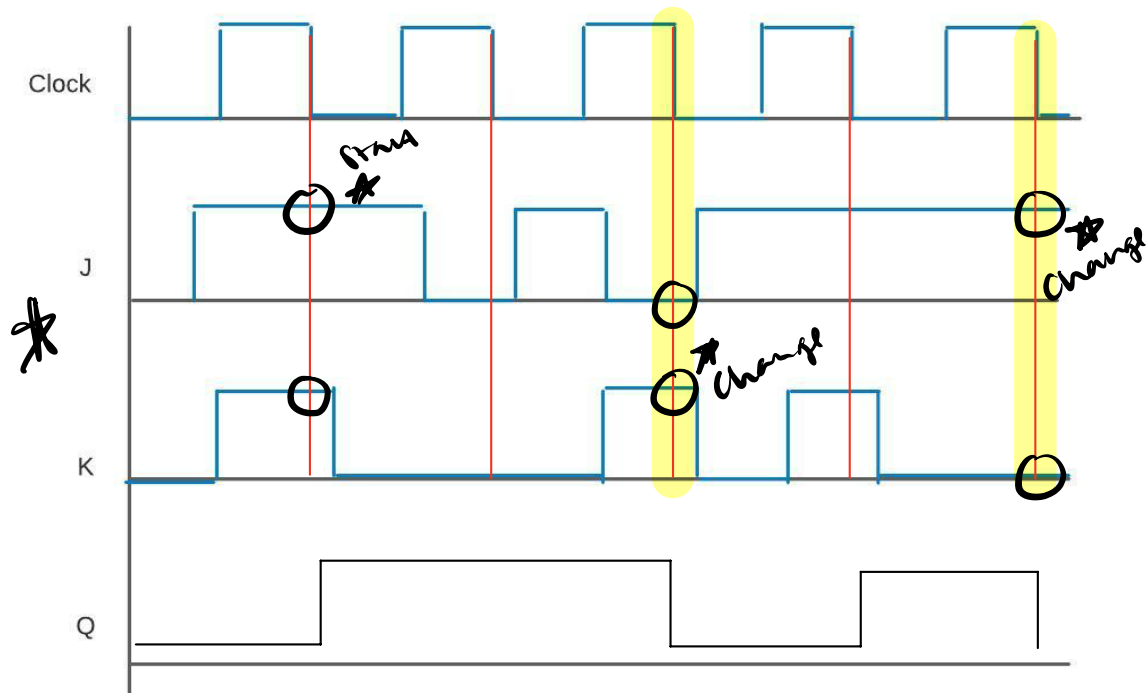
Date DD/MM/YYYY

Question	Points	Maximum
1		24
2		24
3		24
4		24
5		24

Total: _____/100 possible points + 20 Bonus

Question 1:

- Convert a T flip flop to a J-K flip flop using external gates. Explain reasoning (18 points)
- Complete the timing diagram for the falling edge triggered J-K flip flop. (Assume Q begins at 1) (6 points)



T	Q	Q+	J _T	K _T
0	0	0	0	X
0	1	1	X	0
1	0	1	1	X
1	1	0	X	1

$$Q+ = T'Q + TQ'$$

$$Q+ = JQ' + K'Q$$

Q	Q+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Mealy
↓
 $\frac{x}{z}$

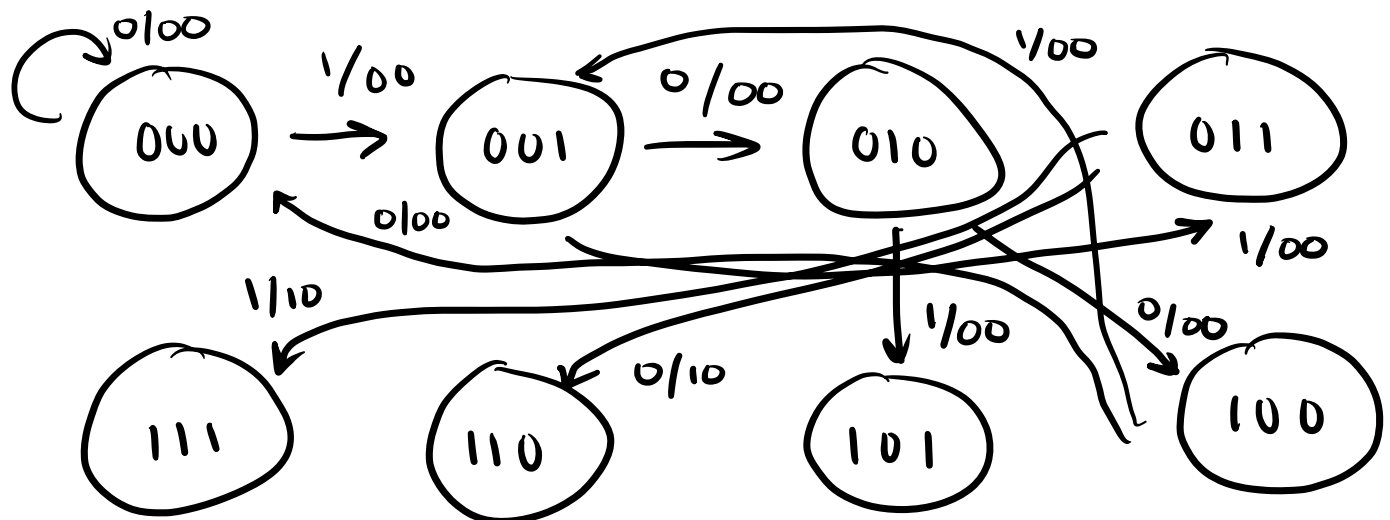
Question 2:

sequence = 1st + 2

Derive the state graph for a Mealy sequential circuit that has one input (X) and two outputs (Z₁ and Z₂). The circuit produces an output of Z₁=1 whenever the sequence 0110 is completed, and an output of Z₂=1 whenever the sequence 1100 is completed. (The minimum number of states is 7)

Example sequence:

X	0	0	1	1	0	1	1	0	1	1	0	1	1	1	0	0	1
Z ₁ (0)	0	0	0	0	1	0	0	0	1	0	0	1	0	0	0	0	0
Z ₂ (0)	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1



A+B+C+			Z ₁ +Z ₂		Z ₁ +Z ₂	
			X=0	X=1	X=0	X=1
S ₀	000	000	00	00	00	00
S ₁	001	010	00	00	00	00
S ₂	010	100	00	00	00	00
S ₃	011	110	00	10	00	00
S ₄	100	000	00	00	00	00
S ₅	101	010	00	00	00	00
S ₆	110	100	00	01	00	00
S ₇	111	110	00	00	00	00

011 = Z₁
110 = Z₂

Question 3:

Reduce the following state table to a minimum number of states.

	Present State	Next State		Output	
		X = 0	X = 1	X = 0	X = 1
s_0	E	H	V	0	1
s_1	D	U	S	0	1
s_2	H	V	E	1	0
s_3	S	U	S	0	1
s_4	U	D	E	1	0
s_5	V	H	E	1	0
s_6	G	U	D	1	0

same s_0 x [s_0 to s_6]
go down (arrow pointing down)
 01
 01
 10
 01
 10
 10
 10

s_1	D	H-U V-S				
s_2	H	X	X			
s_3	S	H-U V-S	X	X		
s_4	U	X	X	V-D	X	
s_5	V	X	X	X	D-H	X
s_6	G	X	X	V-U E-D	D-U E-D	H-U E-D
	E	D	H	S	U	V
	s_0	s_1	s_2	s_3	s_4	s_5

$s_0 + s_6 \neq$ repeat
** NO minimum states **
X = cancel $\rightarrow$ D-V has X
D-V = X

Question 4:

A sequential circuit has one input and one output. The output becomes 1 and remains 1 thereafter when at least one 0 and three 1's have occurred as inputs, regardless of the order of occurrence. Draw the Moore state graph for the circuit. Your final state graph should be neatly drawn with no crossed lines. (The minimum number of states is 8).

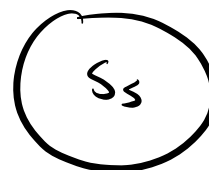
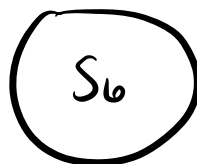
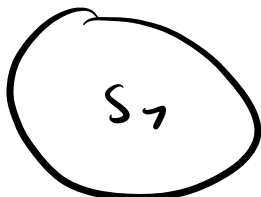
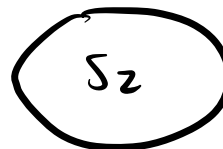
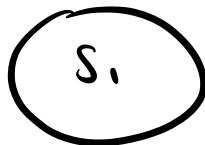
Example sequences: $\uparrow z=1$

X	1	1	1	0	0	0	0
Z	0	0	0	1	1	1	1

X	0	1	0	1	1	0	1
Z	0	0	0	0	1	1	1

0111 $\rightarrow$ 1

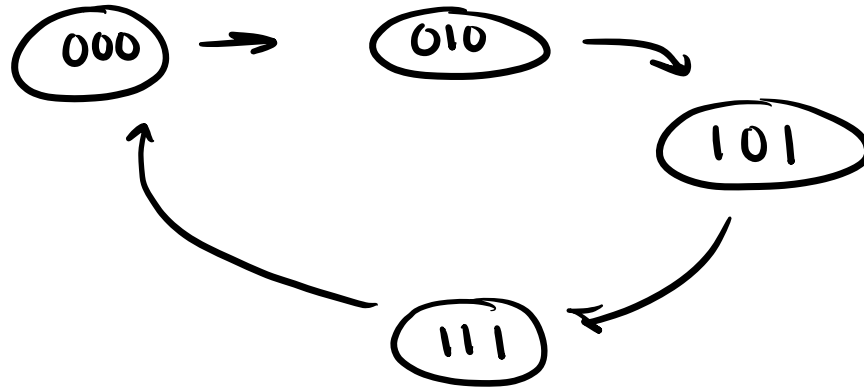
X	0	1	1	0	0	1	1
Z	0	0	0	0	0	1	1



Question 5:

Design a 3-bit counter for the following sequence using T flip-flops:

000, 010, 101, 111, (repeat) 000, 010, ...



Extra Paper